

Math 421

Monday, October 6

Announcements

- Homework 5 due Friday
 - Anything theorems proven or stated in class, readings, homework assignments, and other course related materials, can be cited and used without proof.
- Grace's Drop-In Hours
 - Wednesday: 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Exam 1 – 10/15
 - If you have a conflict, email me (if you haven't already) and **fill out makeup exam request form** (linked from Exam Information page on Canvas) ASAP!

Chapter 2: Sequences

Section 9: Limit Theorems for Sequences

Bounded

Definition. A sequence (s_n) is said to be **bounded** if the set $\{s_n : n \in \mathbb{N}\}$ is a bounded set.

Activity –

1. Give an example of a bounded and an unbounded sequence.
2. Does every bounded sequence converge? Does every unbounded sequence diverge?

Theorem 9.1

Theorem. *Convergent sequences are bounded.*

Discussion.

Proof.

Algebraic Limit Laws

Theorems 9.2-9.6: Let $\lim s_n = s$ and $\lim t_n = t$. Then,

i. $\lim ks_n = ks$ for all $k \in \mathbb{R}$.

ii. $\lim s_n + t_n = s + t$.

iii. $\lim s_n t_n = st$.

iv. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{1}{s_n} = \frac{1}{s}$

v. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$

FALSE Proof of ii. $\lim s_n + t_n = s + t$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists $N > 0$ such that if $n > N$, then $|s_n - s| < \frac{\epsilon}{2}$. Since $\lim t_n = t$, there exists $N > 0$ such that if $n > N$, then $|t_n - t| < \frac{\epsilon}{2}$. Assume $n > N$. Then

$$\begin{aligned} |(s_n + t_n) - (s + t)| &= |(s_n - s) + (t_n - t)| \leq |s_n - s| + |t_n - t| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim s_n + t_n = s + t$.

FALSE Proof of iii. $\lim s_n t_n = st$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists $N_1 > 0$ such that if $n > N_1$, then $|s_n - s| < \frac{\epsilon}{2(|t|+1)}$. Since $\lim t_n = t$, there exists $N_2 > 0$ such that if $n > N_2$, then $|t_n - t| < \frac{\epsilon}{2|s_n|}$. Let $N = \max\{N_1, N_2\}$.

Assume $n > N$. Then

$$\begin{aligned} |s_n t_n - st| &= |s_n t_n - s_n t + s_n t - st| \leq |s_n t_n - s_n t| + |s_n t - st| \\ &\leq |s_n| |t_n - t| + |t| |s_n - s| < |s_n| \frac{\epsilon}{2|s_n|} + |t| \frac{\epsilon}{2(|t|+1)} = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim s_n t_n = st$.

Activity – Prove ii. & iii.

Discussion Questions:

1. What are some common techniques you are seeing in both of these proofs?
2. What are some pitfalls you need to try to avoid?
3. Sketch out a “recipe” to help you structure proofs of this type.

Discussion of Proof of v.

Theorem 9.6: Let $\lim s_n = s$ and $\lim t_n = t$. Then, if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$.

9.7 Theorem (Basic Examples)

- a) $\lim \left(\frac{1}{n^p} \right) = 0$ for $p > 0$.
- b) $\lim a^n = 0$ if $|a| < 1$.
- c) $\lim (n^{1/n}) = 1$.
- d) $\lim (a^{1/n}) = 1$ for $a > 0$.

Diverging to $\pm\infty$

Definition 9.8

- $\lim s_n = +\infty$ provided for each $M > 0$ there is a number N such that $n > N$ implies $s_n > M$.
- $\lim s_n = -\infty$ provided for each $M < 0$ there is a number N such that $n > N$ implies $s_n < M$.

Terminology: A sequence (s_n) has a **limit** or the **limit exists** if (s_n) converges to a real number or diverges to $\pm\infty$.

Example: Show that $\lim n^2 = +\infty$

Activity

1. Prove, directly from the definition, that $\lim(e^n - 7) = +\infty$.
2. Prove that the unbounded sequence $s_n = (-1)^n n$ does not have $-\infty$ or $+\infty$ as a limit.
 - a) What does it mean for an unbounded sequence to not diverge to $+\infty$ or $-\infty$, i.e. what is the negation of the definition?
 - b) What proof technique should you use?

Caution!

The limit laws we've previously stated do not apply to sequences that have infinite limits, we need to prove these separately.

Discussion Question: Can you think of some specific examples where the limit laws for finite limits fail if we tried to apply them to infinite limits?